United States Patent	12406918
Kind Code	B2
Date of Patent	September 02, 2025
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Printed circuit board and electronic component package

Abstract

A printed circuit board includes: a substrate layer in which a plurality of insulating layers and a plurality of wiring patterns are repeatedly layered, the substrate layer including a conductive via layer disposed in one of the plurality of insulating layers to connect wiring patterns, among the plurality of wiring patterns, disposed on upper and lower surfaces of the one insulating layer, respectively; an uppermost substrate layer including an outermost insulating layer disposed outermost within the substrate layer, and a first upper wiring pattern disposed in the outermost insulating layer; and a bump pad disposed on a portion of an upper surface of the first upper wiring pattern and having a length shorter than a length of the first upper wiring pattern.

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Appl. No.: **17/743775**

Filed: **May 13, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230215794 A1	Jul. 06, 2023

Foreign Application Priority Data

KR	10-2022-0000391	Jan. 03, 2022

Publication Classification

Int. Cl.: **H01L23/498** (20060101); **H01L21/48** (20060101); **H01L23/00** (20060101); **H01L23/12** (20060101); **H05K1/11** (20060101); **H05K1/18** (20060101); **H05K3/34** (20060101)

U.S. Cl.:

CPC **H01L23/49838** (20130101); **H01L21/4857** (20130101); **H01L23/49822** (20130101); H01L24/16 (20130101); H01L2224/16227 (20130101); H01L2224/16238 (20130101); H05K2201/09845 (20130101)

Field of Classification Search

CPC: H01L (23/49822); H01L (24/81); H01L (24/13); H01L (24/16); H01L (21/4857); H01L (23/00); H01L (23/49816); H01L (21/48); H05K (2201/09845)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims benefit of priority to Korean Patent Application No. 10-2022-0000391 filed on Jan. 3, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a printed circuit board and an electronic component package including the same.

BACKGROUND

(3) Miniaturization and improvements in performance of small-sized terminal electronic products such as mobile devices are rapidly progressing. Therefore, for a thin and high-performance access point (AP) chip and a non-memory semiconductor chip such as a central processing unit (CPU), a pitch of nodes has been continuously reduced and high integration development has been pursued.

(4) When manufacturing a printed circuit board including bump pads which are densely arranged with a small pitch therebetween and on which a semiconductor chip is mounted, in a case in which a pitch of the bump pads is reduced, a serious short circuit caused by a solder bridge occurs when the chip is mounted, which may be problematic. Further, when soldering is performed, cracking may occur in the bump pad.

(5) In addition, in a case in which solder wicking to the node of the semiconductor chip occurs due to the reduction in the pitch of the bump pads, more serious defects may be caused.

SUMMARY

(6) An aspect of the present disclosure may provide a printed circuit board solving a problem caused when a chip is mounted, even with bump pads densely arranged with a small pitch therebetween.

(7) Another aspect of the present disclosure may provide an electronic component package in which a semiconductor chip having a reduced pitch of nodes is stably mounted on a printed circuit board in which bump pads are densely arranged with a small pitch therebetween.

(8) According to an aspect of the present disclosure, a printed circuit board may include: a substrate layer in which a plurality of insulating layers and a plurality of wiring patterns are repeatedly layered, the substrate layer including a conductive via layer disposed in one of the plurality of insulating layers to connect wiring patterns, among the plurality of wiring patterns, disposed on upper and lower surfaces of the one insulating layer, respectively; an uppermost substrate layer including an outermost insulating layer disposed outermost within the substrate layer, and a first upper wiring pattern disposed in the outermost insulating layer; and a bump pad disposed on a portion of an upper surface of the first upper wiring pattern and having a length shorter than a length of the first upper wiring pattern.

(9) According to another aspect of the present disclosure, a printed circuit board may include: a substrate layer in which a plurality of insulating layers and a plurality of wiring patterns are repeatedly layered, the substrate layer including a conductive via layer disposed in one of the plurality of insulating layers to connect wiring patterns, among the plurality of wiring patterns, disposed on upper and lower surfaces of the one insulating layer, respectively; an uppermost substrate layer including an outermost insulating layer disposed outermost within the substrate layer, and a first upper wiring pattern disposed in the outermost insulating layer; a bump pad disposed on a portion of an upper surface of the first upper wiring pattern; and a solder resist layer having a hole exposing the bump pad. Another portion of the upper surface of the first upper wiring pattern may be lower than an upper surface of the outermost insulating layer, the outermost insulating layer may have a solder dam protruding so as to be higher than the one portion of the upper surface of the first upper wiring pattern, and a recess portion is provided between a side surface of the solder dam and a side surface of the bump pad.

(10) According to another aspect of the present disclosure, an electronic component package may

include: a printed circuit board including a substrate layer in which a plurality of insulating layers and a plurality of wiring patterns are repeatedly layered, the substrate layer including a conductive via layer disposed in one of the plurality of insulating layers to connect wiring patterns, among the plurality of wiring patterns, disposed on upper and lower surfaces of the one insulating layer, respectively, an uppermost substrate layer including an outermost insulating layer disposed outermost within the substrate layer, and a first upper wiring pattern disposed in the outermost insulating layer, and a bump pad disposed on a portion of an upper surface of the first upper wiring pattern and having a length shorter than a length of the first upper wiring pattern; and a solder resist layer having a hole exposing the bump pad; and a semiconductor chip connected to the bump pad by a solder.
(11) According to another aspect of the present disclosure, a printed circuit board may include: a substrate layer in which a plurality of insulating layers and a plurality of wiring patterns are repeatedly layered, the substrate layer including a conductive via layer disposed in one of the plurality of insulating layers to connect wiring patterns, among the plurality of wiring patterns, disposed on upper and lower surfaces of the one insulating layer, respectively; an uppermost substrate layer including an outermost insulating layer disposed outermost within the substrate layer, and a first upper wiring pattern disposed in the outermost insulating layer; and a bump pad disposed on the first upper wiring pattern. The outermost insulating layer may include a dam protruding with respect to a recess portion disposed between the outermost insulating layer and the bump pad.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a block diagram schematically illustrating an example of an electronic device system;
(3) FIG. **2** is a perspective view schematically illustrating an example of an electronic device;
(4) FIG. **3** is a cross-sectional view schematically illustrating a cross section of a printed circuit board according to an exemplary embodiment in the present disclosure;
(5) FIG. **4** is an enlarged view of Part A of FIG. **3**;
(6) FIG. **5** is an enlarged view of Part B of FIG. **4**;
(7) FIG. **6** is an enlarged view of Part C of FIG. **4**;
(8) FIGS. **7***a* through **7***d* are cross-sectional views illustrating a process of manufacturing the printed circuit board before separation from a carrier in a method for manufacturing the printed circuit board of FIG. **3**;
(9) FIGS. **8***a* through **8***c* are cross-sectional views illustrating a process of manufacturing the printed circuit board after separation from the carrier in the method for manufacturing the printed circuit board of FIG. **3**; and
(10) FIG. **9** is a schematic cross-sectional view of an electronic component package according to an exemplary embodiment in the present disclosure.
DETAILED DESCRIPTION
(11) Hereinafter, exemplary embodiments in the present disclosure will now be described in detail with reference to the accompanying drawings.
(12) Electronic Device System
(13) FIG. **1** is a block diagram schematically illustrating an example of an electronic device system.
(14) Referring to FIG. **1**, an electronic device **1000** may accommodate a main board **1010** therein. The main board **1010** may include chip-related components **1020**, network-related components

1030, other components **1040**, and the like, physically or electrically connected thereto. These components may be connected to others to be described below to form various signal lines **1090**.
(15) The chip-related components **1020** may include a memory chip such as a volatile memory (for example, a dynamic random access memory (DRAM)), a non-volatile memory (for example, a read only memory (ROM)), or a flash memory; an application processor chip such as a central processor (for example, a central processing unit (CPU)), a graphics processor (for example, a graphics processing unit (GPU)), a digital signal processor, a cryptographic processor, a microprocessor, or a microcontroller; and a logic chip such as an analog-to-digital (ADC) converter or an application-specific integrated circuit (ASIC). However, the chip-related components **1020** are not limited thereto, but may also include other types of chip-related components. Further, these chip-related components may be combined with each other. The chip-related components **1020** may be in a form of a package including the above-described chips.
(16) The network-related components **1030** may include protocols such as wireless fidelity (Wi-Fi) (Institute of Electrical And Electronics Engineers (IEEE) 802.11 family, or the like), worldwide interoperability for microwave access (WiMAX) (IEEE 802.16 family, or the like), IEEE 802.20, long term evolution (LTE), evolution data only (Ev-DO), high speed packet access+(HSPA+), high speed downlink packet access+(HSDPA+), high speed uplink packet access+(HSUPA+), enhanced data GSM environment (EDGE), global system for mobile communications (GSM), global positioning system (GPS), general packet radio service (GPRS), code division multiple access (CDMA), time division multiple access (TDMA), digital enhanced cordless telecommunications (DECT), Bluetooth, 3G, 4G, and 5G protocols, and any other wireless and wired protocols, designated after the abovementioned protocols. However, the network-related components **1030** are not limited thereto, but may also include a variety of other wireless or wired standards or protocols. In addition, the network-related components **1030** may be combined with the chip-related components **1020** and provided in a form of a package.
(17) Other components **1040** may include a high frequency inductor, a ferrite inductor, a power inductor, ferrite beads, a low temperature co-fired ceramic (LTCC), an electromagnetic interference (EMI) filter, a multilayer ceramic capacitor (MLCC), or the like. However, other components **1040** are not limited thereto, and may also include passive elements in a form of a chip component used for various other purposes, or the like. In addition, other components **1040** may be combined with the chip-related components **1020** and/or network-related components **1030** and provided in a form of a package.
(18) Depending on a type of the electronic device **1000**, the electronic device **1000** may include other components that may or may not be physically or electrically connected to the main board **1010**. Examples of other components include a camera module **1050**, an antenna module **1060**, a display **1070**, and a battery **1080**. However, other components are not limited thereto, but may include an audio codec, a video codec, a power amplifier, a compass, an accelerometer, a gyroscope, a speaker, a mass storage device (for example, a hard disk drive), a compact disk (CD), a digital versatile disk (DVD), and the like. In addition, the electronic device **1000** may include other components used for various purposes depending on the type of the electronic device **1000**.
(19) The electronic device **1000** may be a smartphone, a personal digital assistant (PDA), a digital video camera, a digital still camera, a network system, a computer, a monitor, a tablet PC, a laptop PC, a netbook PC, a television, a video game machine, a smartwatch, an automotive component, or the like. However, the electronic device **1000** is not limited thereto, but may be any other electronic device processing data.
(20) FIG. **2** is a perspective view schematically illustrating an example of the electronic device.
(21) Referring to FIG. **2**, the electronic device may be, for example, a smartphone **1100**. A motherboard **1110** may be accommodated in the smartphone **1100**, and various electronic components **1120** are physically and/or electrically connected to the motherboard **1110**. Further, a camera module **1130**, a speaker **1140**, and/or the like are accommodated in the smartphone **1000**.

Some of the electronic components **1120** may be the above-described chip-related components such as a bridge-embedded board **1121** having a surface on which a plurality of electronic components are mounted, but the electronic components **1120** are not limited thereto. Meanwhile, the electronic device is not necessarily limited to the smartphone **1100**, but may be other electronic devices as described above.

(22) Printed Circuit Board

(23) FIG. **3** is a cross-sectional view schematically illustrating a cross section of a printed circuit board according to an exemplary embodiment in the present disclosure, FIG. **4** is an enlarged view of Part A of FIG. **3**, FIG. **5** is an enlarged view of Part B of FIG. **4**, and FIG. **6** is an enlarged view of Part C of FIG. **4**.

(24) Referring to FIGS. **3** through **6**, a printed circuit board **1** according to an exemplary embodiment in the present disclosure may include substrate layers **10**, **20**, and **30**, conductor via layers **15**, **25**, and **35**, an uppermost substrate layer **10**, an external connection pad **60**, and a bump pad **62**.

(25) Each of the substrate layers **10**, **20**, and **30** may include a plurality of insulating layers **12**, **22**, and **32** and a plurality of wiring patterns **14**, **24**, and **34**. In the printed circuit board **1**, the plurality of insulating layers **12**, **22**, and **32** and the plurality of wiring patterns **14**, **24**, and **34** may be repeatedly layered.

(26) In a case in which the substrate layers in the printed circuit board **1** are referred to as a first substrate layer **10**, a second substrate layer **20**, and a third substrate layer **30**, respectively, from the top in the drawing, a first wiring pattern **14** in an upper surface of a first insulating layer **12** in the first substrate layer **10**, and a second wiring pattern **24** on a lower surface of the first insulating layer **12** may be connected to each other by a first conductive via layer **15** penetrating through the first insulating layer **12**.

(27) In this way, the second wiring pattern **24** and a third wiring pattern **34** may be connected to a second conductor via layer **25**, and the third wiring pattern **34** and a lowermost external connection pad **44** may be connected to the third conductive via layer **35**.

(28) According to the exemplary embodiment illustrated in FIG. **3**, in the printed circuit board **1**, each of the number of insulating layers **12**, **22**, and **32** and the number of wiring patterns **14**, **24**, and **34** is three, but the printed circuit board **1** may include more built-up insulating layers and wiring patterns as necessary.

(29) The insulating layers **12**, **22**, and **32** may be formed of a glass-containing insulating material or a glass-free inorganic insulating resin. Prepreg (PPG) may be used as the glass-containing insulating material, and an Ajinomoto build-up film (ABF) may be used as the glass-free inorganic insulating resin. However, the insulating material is not particularly limited.

(30) In addition, the insulating layers **12**, **22**, and **32** may be organic insulating layer including at least one of the ABF and polyimide, but are not limited thereto.

(31) Among the wiring patterns **14**, **24**, and **34**, a wiring pattern formed in the outermost layer of the printed circuit board **1** and exposed to the outside may be referred to as a connection pad. The connection pad may be connected to an external semiconductor chip or another board.

(32) The wiring patterns **14**, **24**, and **34** may be conductive metal layers formed by electroplating, chemical plating, or sputtering, and materials of the wiring patterns **14**, **24**, and **34** may be copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or alloys thereof.

(33) A metal material may be used as a material of the plurality of conductive via layers **15**, **25**, and **35**, and examples of the metal material may include copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), and an alloy thereof. The plurality of conductor via layers **15**, **25**, and **35** may include a signal connection via, a ground connection via, a power connection via, and the like according to the design. Each of wiring vias of the plurality of conductor via layers **15**, **25**, and **35** may be completely filled with a metal material, or the metal

material may be formed along a wall surface of a via hole. The plurality of conductive via layers **15**, **25**, and **35** may have a tapered shape, and may be formed by plating, for example, an additive process (AP), a semi additive process (SAP), a modified semi additive process (MSAP), tenting (TT), or the like. As a result, each of the plurality of conductive via layers **15**, **25**, and **35** may include a seed layer, which is an electroless plating layer, and an electrolytic plating layer formed based on the seed layer. The number of conductor via layers **15**, **25**, and **35** may be more or less than illustrated in the drawings.

(34) Here, in a case in which the first substrate layer **10** formed at the uppermost portion of the substrate layers **10**, **20**, and **30** is referred to as the uppermost substrate layer, a first upper wiring pattern **14-1** and a second upper wiring pattern **14-2** may be disposed in the outermost insulating layer **12**. The first upper wiring pattern **14-1** may be disposed to be spaced apart from the second upper wiring pattern **14-2**, the external connection pad **60** may be disposed on an upper surface of the second wiring pattern **14-2**, and the bump pad **62** may be disposed on an upper surface of the first upper wiring pattern **14-1**. A length l.sub.p of the external connection pad **60** may be larger than a length l.sub.b of the bump pad **62**.

(35) The bump pad **62** may be an external connection pad for mounting a semiconductor chip in which nodes are formed with a small pitch therebetween, and the external connection pad **60** may be a wiring pattern to which another external printed circuit board, a passive component, or the like is connected by soldering.

(36) A pitch between the bump pads **62** may be small, and the bump pads **62** may be densely connected to the nodes of the semiconductor chip.

(37) Solder resist layers **52** and **54** may be formed on the outermost layer of the printed circuit board **1** to prevent reflow of a solder at the time of connection to an external electronic component. The solder resist layers **52** and **54** may have holes **55** exposing the external connection pad **60** and the bump pad **62**. In one example, the solder resist layer **54** may be spaced apart from the bump pad **62**.

(38) The solder resist layers **52** and **54** may be formed of a photosensitive material or a thermosetting material, and may be formed by a screen printing method. In this case, a solder resist **50** may be formed by, for example, laminating or coating a film-shaped resist material.

(39) Referring to FIGS. **4** and **5**, a length l.sub.2 of the second upper wiring pattern **14-2** may be smaller than the length l.sub.p of the external connection pad **60**. Since the length l.sub.p of the external connection pad **60** is larger than that of the second upper wiring pattern **14-2**, mounting strength may increase in surface mounting. In addition, a height of a solder ball may be higher in a case in which another wiring layer, that is, the external connection pad **60**, is provided, than in a case in which the second upper wiring pattern **14-2** serves as the external connection pad, and since a thickness of a central portion of the external connection pad **60** is increased, solder ball crack defects caused by metal consumption of the connection pad may be suppressed.

(40) FIGS. **4** and **6** illustrate the bump pads **62** arranged with a small pitch therebetween.

(41) The length l.sub.b of the bump pad **62** is smaller than a length l.sub.1 of the first upper wiring pattern **14-1**. An upper surface of the bump pad **62** is higher than an upper surface of the outermost insulating layer **12**. In addition, the upper surface of the first upper wiring pattern **14-1** that is not covered by the bump pad **62** is lower than the upper surface of the outermost insulating layer **12**, and the outermost insulating layer **12** has a solder dam **12**D protruding so as to be higher than the upper surface of the first upper wiring pattern **14-1**.

(42) A space **80** (e.g., a recess portion or a groove portion) may be formed between a side surface of the solder dam **12**D and a side surface of the bump pad **62** to accommodate a solder during soldering, thereby effectively preventing a short circuit caused by a solder bridge from occurring between the bump pads **62**. A length of the space **80** between the side surface of the solder dam **12**D and the side surface of the bump pad **62** may be 5 μm or more, which is effective in preventing the short circuit caused by a solder bridge.

(43) Since the solder dam **12**D is formed when a seed metal foil of a carrier is removed during a process of manufacturing of the printed circuit board, the solder dam **12**D has a naturally determined depth, and it is preferable that a depth of a step of the solder dam **12**D is 1 to 6 μm.
(44) In general, in a case in which the pitch between the bump pads **62** is 90 μm or less, a risk of soldering failure is high when soldering, but the bump pad **62** having a structure according to the present disclosure is effective in preventing a short circuit caused by solder bleeding.
(45) In a case in which a height from the upper surface of the insulating layer **12** to the upper surface of the bump pad **62** higher than the upper surface of the insulating layer **12** is defined as a bump pad protrusion height h.sub.b, when the bump pad protrusion height h.sub.b is 7 μm or more, the solder ball crack defects due to metal consumption of the bump pad **62** may be suppressed, and a height of the package may be effectively reduced. When the bump pad protrusion height h.sub.b is less than 7 μm, the solder ball crack defects due to metal consumption of the bump pad **62** may not be effectively suppressed.
(46) Meanwhile, since there is no other wiring pattern passing between the bump pad **62** and another bump pad **64**, a short circuit caused by a solder bridge may be prevented from occurring between the bump pad **62** and another bump pad **64** arranged with a small pitch therebetween. FIGS. **7***a* through **7***d* are cross-sectional views illustrating a process S**1** of manufacturing the printed circuit board before separation from the carrier in a method for manufacturing the printed circuit board of FIG. **3**.
(47) FIG. **7***a* is a cross-sectional view of a state in which a carrier **100** is provided and the first wiring pattern **14** is formed. The carrier refers to a structure capable of carrying a board, and the carrier **100** may include, for example, a rigid structure formed of silicon or a glass wafer or containing a metal. A bonding layer **102** and a metal foil **105** serving as a seed layer for forming the wiring pattern may be formed on the carrier **100**. The metal foil **105** may be a conductive copper foil.
(48) Although the process for the printed circuit board may be performed on both upper and lower sides of the carrier **100**, only the process performed on one side of the carrier is illustrated for the sake of simplification of the drawing.
(49) FIG. **7***b* is a cross-sectional view illustrating a process S**2** of building up the outermost insulating layer **12**, the first conductive via layer **15** in the outermost insulating layer **12**, and the second wiring pattern **24** on the outermost insulating layer **12**, so that the first conductive via layer **15** connects the first wiring pattern **14** and the second wiring pattern **24**. Further, FIG. **7***c* is a cross-sectional view illustrating a process S**3** of building up the second insulating layer **22**, the second conductive via layer **25** in the second insulating layer **22**, and the third wiring pattern **34** on the second insulating layer **22**, so that the second conductive via layer **25** connects the second wiring pattern **24** and the third wiring pattern **34**.
(50) More insulating layers, wiring patterns, and conductive via layers may be built up as needed.
(51) FIG. **7***d* illustrates a process S**4** of separating the built-up substrate layer from the carrier. Here, the metal foil **105** attached to the carrier **100** may be separated together with the substrate layer.
(52) FIGS. **8***a* through **8***c* are cross-sectional views illustrating a process of manufacturing the printed circuit board after separation from the carrier in the method for manufacturing the printed circuit board of FIG. **3**.
(53) As illustrated in FIG. **8***a*, the external connection pad **60** and the bump pad **62** may be formed on the separated substrate layer by using the metal foil **105** as the seed layer, and the external connection pad **44** may be formed on a lower surface of the lowermost insulating layer **32** of the substrate layer.
(54) FIG. **8***b* illustrates a process for removing the metal foil **105** by etching. Here, portions of the metal foil **105** that are covered by the external connection pad **60** and the bump pads **62** and **64** are maintained and are not removed, but portions of the metal foil **105** that are not covered by the

external connection pad **60** and the bump pads **62** and **64** may be removed by etching. When the portions of the metal foil **105** that are not covered by the bump pads **62** and **64** are etched, in a case in which a component under the metal foil **105** is not the insulating layer **12**, but the first upper wiring pattern **14-1**, the first upper wiring pattern **14-1** may be partially etched together, such that the solder dam **12**D is formed in the insulating layer **12**.
(55) The depth of the solder dam **12**D preventing a short circuit caused by a solder bridge in the insulating layer **12** is determined by the above-described etching.
(56) FIG. **8***c* illustrates a process of applying the solder resists **52** and **54** and forming the holes **55** to expose the external connection pads **60** and **44** and the bump pads **62** and **64** to the outside.
(57) Electronic Component Package
(58) FIG. **9** is a schematic cross-sectional view of an electronic component package according to an exemplary embodiment in the present disclosure.
(59) Referring to FIG. **9**, an electronic component package **500** according to an exemplary embodiment in the present disclosure may include the above-described printed circuit board **1** and a semiconductor chip **300**.
(60) The printed circuit board **1** may include the external connection pad **60** and the bump pads **62** and **64** as described above. Semiconductor nodes **350** may be connected to the densely formed bump pads **62** and **64**, respectively, by soldering **252**. A solder ball **250** of the external connection pad **60** may be coupled to another external printed circuit board or a passive component.
(61) In addition, the semiconductor chip **300** may be molded with an encapsulation layer and may thus be protected from an external environment.
(62) As set forth above, according to the exemplary embodiment in the present disclosure, with the printed circuit board according to the present disclosure, problems such as a short circuit caused by a solder bridge and a warpage in a packaging process caused when the chip is mounted even with the bump pads densely arranged with a small pitch therebetween may be solved.
(63) In the printed circuit board according to the present disclosure, for example, the external connection pad and the bump pad that have different lengths may be simultaneously formed, which may improve process efficiency.
(64) With the printed circuit board according to the present disclosure, the soldering performance is improved even when the circuit board has a small size, such that the overall packaging performance may be improved.
(65) While exemplary embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A printed circuit board comprising: a substrate layer in which a plurality of insulating layers and a plurality of wiring patterns are repeatedly layered in a thickness direction, the substrate layer including a conductive via layer disposed in one of the plurality of insulating layers to connect wiring patterns, among the plurality of wiring patterns, disposed on upper and lower surfaces of the one of the plurality of insulating layers, respectively; an uppermost substrate layer including an outermost insulating layer disposed on an outermost surface of the substrate layer, and a first upper wiring pattern disposed in the outermost insulating layer; a bump pad disposed on a portion of an upper surface of the first upper wiring pattern and having a length shorter than a length of the first upper wiring pattern; and a solder resist layer having a first hole exposing the bump pad, and having a height greater than that of the bump pad with respect to an upper surface of the outermost insulating layer, wherein in a cross section of the printed circuit board cut in the thickness direction, a side surface of the bump pad extends in the thickness direction, another portion of the upper surface of the first upper wiring pattern is lower than the upper surface of the outermost

insulating layer, and the outermost insulating layer has a solder dam protruding so as to be higher than the another portion of the upper surface of the first upper wiring pattern.
2. The printed circuit board of claim 1, wherein an upper surface of the bump pad is higher than the upper surface of the outermost insulating layer.
3. The printed circuit board of claim 1, wherein a recess portion is provided between a side surface of the solder dam and a side surface of the bump pad.
4. The printed circuit board of claim 1, wherein the uppermost substrate layer includes a second upper wiring pattern disposed to be spaced apart from the first upper wiring pattern, an external connection pad is disposed on an upper surface of the second upper wiring pattern, and a length of the external connection pad is larger than a length of the second upper wiring pattern.
5. The printed circuit board of claim 4, wherein the solder resist layer covers a portion of the external connection pad and has a second hole exposing another portion of the external connection pad, and the height of the solder resist layer is greater than that of the external connection pad with respect to the upper surface of the outermost insulating layer.
6. The printed circuit board of claim 5, wherein the solder resist layer includes a photosensitive material or a thermosetting material.
7. A printed circuit board comprising: a substrate layer in which a plurality of insulating layers and a plurality of wiring patterns are repeatedly layered in a thickness direction, the substrate layer including a conductive via layer disposed in one of the plurality of insulating layers to connect wiring patterns, among the plurality of wiring patterns, disposed on upper and lower surfaces of the one of the plurality of insulating layers, respectively; an uppermost substrate layer including an outermost insulating layer disposed on an outermost surface of the substrate layer, and a first upper wiring pattern disposed in the outermost insulating layer; a bump pad disposed on a portion of an upper surface of the first upper wiring pattern; and a solder resist layer having a hole exposing the bump pad, and having a height greater than the bump pad with respect to an upper surface of the outermost insulating layer, wherein another portion of the upper surface of the first upper wiring pattern is lower than the upper surface of the outermost insulating layer, the outermost insulating layer has a solder dam protruding so as to be higher than the another portion of the upper surface of the first upper wiring pattern, a recess portion is provided between a side surface of the solder dam and a side surface of the bump pad, in a cross section of the printed circuit board cut in the thickness direction, a side surface of the bump pad extends in the thickness direction, the uppermost substrate layer includes a second upper wiring pattern disposed to be spaced apart from the first upper wiring pattern, an external connection pad is disposed on an upper surface of the second upper wiring pattern, and a length of the external connection pad is larger than a length of the second upper wiring pattern.
8. The printed circuit board of claim 7, wherein an upper surface of the bump pad is higher than the upper surface of the outermost insulating layer.
9. The printed circuit board of claim 7, wherein the solder resist layer includes a photosensitive material or a thermosetting material.
10. An electronic component package comprising: a printed circuit board including a substrate layer in which a plurality of insulating layers and a plurality of wiring patterns are repeatedly layered, the substrate layer including a conductive via layer disposed in one of the plurality of insulating layers to connect wiring patterns, among the plurality of wiring patterns, disposed on upper and lower surfaces of the one of the plurality of insulating layers, respectively, an uppermost substrate layer including an outermost insulating layer disposed on an outermost surface of the substrate layer, and a first upper wiring pattern disposed in the outermost insulating layer, and a bump pad disposed on a portion of an upper surface of the first upper wiring pattern and having a length shorter than a length of the first upper wiring pattern, and a solder resist layer having a hole exposing the bump pad and having a height greater than that of the bump pad with respect to an upper surface of the outermost insulating layer; and a semiconductor chip connected to the bump

pad by a solder different from the bump pad, wherein an upper surface of the bump pad is higher than the upper surface of the outermost insulating layer, another portion of the upper surface of the first upper wiring pattern is lower than the upper surface of the outermost insulating layer, and the outermost insulating layer has a solder dam protruding so as to be higher than the another portion of the upper surface of the first upper wiring pattern.

11. The electronic component package of claim 10, wherein a recess portion is provided between a side surface of the solder dam and a side surface of the bump pad.

12. The electronic component package of claim 10, wherein the uppermost substrate layer includes a second upper wiring pattern disposed to be spaced apart from the first upper wiring pattern, an external connection pad is disposed on an upper surface of the second upper wiring pattern, and a length of the external connection pad is larger than a length of the second upper wiring pattern.

13. A printed circuit board comprising: a substrate layer in which a plurality of insulating layers and a plurality of wiring patterns are repeatedly layered in a thickness direction, the substrate layer including a conductive via layer disposed in one of the plurality of insulating layers to connect wiring patterns, among the plurality of wiring patterns, disposed on upper and lower surfaces of the one of the plurality of insulating layers, respectively; an uppermost substrate layer including an outermost insulating layer disposed on an outermost surface of the substrate layer, and a first upper wiring pattern disposed in the outermost insulating layer; a bump pad disposed on the first upper wiring pattern; and a solder resist layer having a hole exposing the bump pad, and having a height greater than that of the bump pad with respect to an upper surface of the outermost insulating layer, wherein the outermost insulating layer includes a dam protruding with respect to a recess portion disposed between the outermost insulating layer and the bump pad, in a cross section of the printed circuit board cut in the thickness direction, a side surface of the bump pad extends in the thickness direction, the uppermost substrate layer includes a second upper wiring pattern spaced apart from the first upper wiring pattern, and an external connection pad is disposed on an upper surface of the second upper wiring pattern.

14. The printed circuit board of claim 13, wherein an upper surface of the bump pad is higher than the upper surface of the outermost insulating layer.

15. The printed circuit board of claim 13, wherein the bump pad has a length shorter than a length of the first upper wiring pattern, and the external connection pad has a length larger than a length of the second upper wiring pattern.

16. The printed circuit board of claim 13, wherein the solder resist layer is spaced apart from the bump pad, covers a portion of the external connection pad, and has another hole exposing another portion of the external connection pad, and the height of the solder resist layer is greater than that of the external connection pad with respect to the upper surface of the outermost insulating layer.

17. The printed circuit board of claim 13, wherein the recess portion surrounds the bump pad.

18. The printed circuit board of claim 13, wherein the recess portion is provided in the first upper wiring pattern.